The Ganong effect is sensitive to noise

Chang Wang[1], Stephen Politzer-Ahles[1]

[1]*University of Kansas*

Introduction. Ambiguous sounds tend to be judged in a way such that they yield a real word in their context, all else being equal [Ganong, 1980]. For example, given a sound ambiguous between [d] and [t], English speakers are more likely to judge that sound as being a /t/ if it's in the stimulus _*ask* (where *task* is a real word and **dask* is not) than in the stimulus _*esk* (where **tesk* is not a real word, but *desk* is).

Accounts of the source of this effect make different predictions about how it should interact with various factors, including stimulus quality. The extent to which impoverishment of the bottom-up signal (e.g., through presenting stimuli in noise) increases or decreases the effect of lexical context on phonetic identification could help adjudicate between different models of phoneme recognition. However, the literature has not conclusively demonstrated whether noise affects the size of the lexical effect. A series of studies several decades ago, mostly with small samples and several with noise only manipulated between participants, found inconsistent effects that could not easily be explained by moderating factors (Burton et al., 1989; Burton & Blumstein, 1995; McQueen, 1991, 2003; Pitt & Samuel, 1993), and to our knowledge the issue hasn't been investigated since.

In the present study, we test whether the Ganong effect is increased by noise, using a large sample size, modern statistical methods for accounting for differences in response time, a larger set of items, and a novel noise manipulation. (Previous studies used noise similar to white noise, which may intrinsically overlap with aspiration noise and thus may shift phonetic identification independently of any lexical effects. The present study instead uses multi-talker babble.)

Methods. We selected nine Mandarin disyllabic words with unaspirated onsets (three labial, three alveolar, three velar) and nine corresponding disyllabic words with aspirated onsets; see Table 1. We created continua from unaspirated to aspirated by starting from aspirated tokens and successively cutting out more and more aspiration to create tokens with shorter and shorter voice onset times (VOTs). For the experiment we used tokens with {5, 15, 20, 25, 30, 65}-ms VOTs, based on pilot identifications showing 15-30ms tokens to be fairly ambiguous.

We created multi-talker babble by having four Mandarin speakers (2M, 2F) read self-selected texts aloud. For each critical stimulus, we randomly selected a corresponding duration of speech from each of the four speakers, intensity-normalized them, and combined them into one audio file with the token, with an SNR of 5 dB (stimuli were 65 dB and noise was 60 dB).

Data were collected from 60 Mandarin speakers online using the Gorilla experiment control platform (www.gorilla.sc; Anwyl-Irvine et al., 2019). Each participant heard 2 repetitions of each token with and 2 without, yielding 384 observations per participant (2 repetitions * 6 steps * 16 continua * 2 noise continua).

Results. Strong Ganong effects were observed both without noise and with noise (Figure 1). Consistent with many previous studies, the size of the Ganong effect was largest when participants responded quickly (Figure 2, following Mattys & Wiget, 2011). A Bayesian generalized linear mixed-effects model revealed both a bias*noise interaction (larger Ganong effect in noise; 95% CrI: [0.03, 0.49]) and a bias*RT interaction (smaller Ganong effect in higher RTs; 95% CrI: [-0.57, -0.10]).

Discussion. The Ganong lexical bias effect increased reliably, but subtly, when stimuli were presented in noise. The present study had greater power, more appropriate statistics, and a more generalizable set of items than early investigations of this effect did. The fact that noise moderates this effect can help resolve debates over the cognitive locus of the effect.

Table 1. Continua used in the experiment

Bias	Labial items	Alveolar items	Velar items
Unaspirated is real word	*bìmiǎn – *pìmiǎn* *bìxū – *pìxū* *bìzhi – *pìzhi*	*daibiao – *taibiao* *daiti – *taiti* *daijia – *taijia*	*gongchi – *kongchi* *gongfu – *kongfu* *gonghui – *konghui*
Aspirated is real word	**bìru – pìru* **bìgu – pìgu* **bìmei – pìmei*	**daiji – taiji* **daidu – taidu* **daikong – taikong*	**gongdong – kongdong* **gongfu – kongfu* **gongtiao – kongtiao*

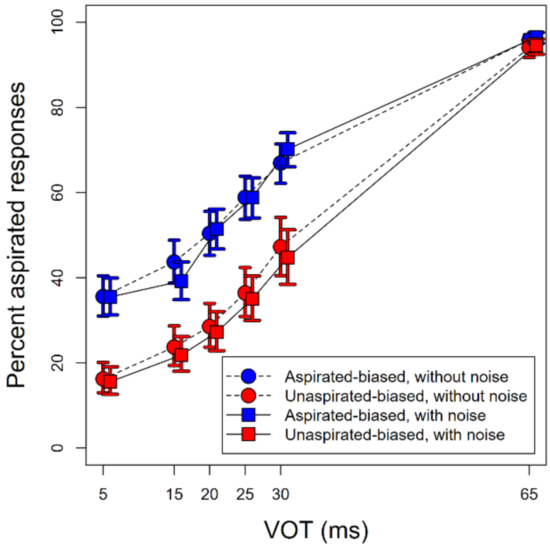

Fig. 1. Ganong effect (more "Aspirated" judgments in aspirated-biasing contexts) as a function of VOT step. Error bars represent 95% confidence intervals from a glmer() model.

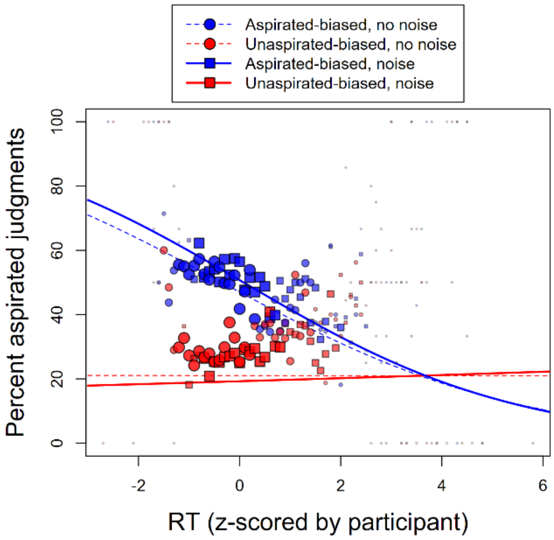

Fig. 2. Ganong effect (averaged over 5-30ms VOTs – excluding 65ms because this VOT value is unambiguous and thus does not elicit a Ganong effect) as a function of reaction time (normalized per participant). Points represent averages over 0.1-unit RT bins (larger/darker points represent bins with more observations). Lines are predictions from a Bayesian GLMM.

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